

AI PROJECT BRIEF

Homework Guide + Fill-In Template

Define a useful AI workflow before building a skill, agent, plugin, or app.

Student name	Course / section	Due date
<i>Click or type here</i>	<i>Click or type here</i>	<i>Click or type here</i>

Estimated completion time: 45–75 minutes

Submit this completed document unless your instructor provides a different format.

START

The main idea: Do not begin by deciding to build “an AI app.” Begin with a specific person, a repeated problem, and a workflow that produces a useful result.

Your Assignment

Create a one-page project brief for an AI workflow you would genuinely want to build. The brief should explain the user, the problem, the current process, the proposed AI workflow, and how you will know it is useful.

GOAL

Your goal: By the end of this homework, another person should be able to understand exactly what your project will do—without seeing a demo or asking you to explain it.

What you are designing

Your first version is a **workflow**: a repeatable sequence in which Claude turns defined inputs into useful outputs. Later, you may package that workflow as a Claude skill, an agent, a plugin, or an app.



What to submit

- A completed Project Brief using the template in this document.
- A one-sentence project definition.
- One realistic example of the input your project will receive.
- One realistic example of the output it should produce.
- A short explanation of what the first version will not do.

How to use Claude while completing this homework

- Use Claude to brainstorm possible problems, challenge your assumptions, and improve clarity.
- Do not ask Claude to invent a problem you do not understand. Choose a task you have experienced, observed, or can research responsibly.
- Review every suggestion. Your final brief should reflect your own judgment.

OPTIONAL CLAUDE PROMPT

“Act as an AI project coach. Ask me one question at a time to help me identify a specific user, a repeated problem, the current workflow, and a realistic AI-assisted workflow. Push back if my idea is too broad or if AI is unnecessary.”

1

Choose a Problem Worth Solving

Start with a real task—not a technology or a broad industry.

Strong first projects usually improve work that is repetitive, information-heavy, inconsistent, slow, or difficult to personalize. They do not need to be revolutionary. They need to be useful and testable.

Stronger starting point	Too broad or unclear
Turn class notes into a study guide and practice quiz.	Build an AI education platform.
Compare a résumé with one job description and identify evidence gaps.	Create an AI career coach.
Summarize customer feedback and group repeated complaints.	Use AI for business.
Convert meeting notes into decisions, owners, and deadlines.	Build a productivity agent.
Research one company and prepare interview questions.	Build a research app.

Use the “one user, one problem” rule

- One primary user: the person who directly uses or benefits from the workflow.
- One main problem: a specific job they are trying to complete.
- One starting input: the information that begins the workflow.
- One meaningful output: the result the user can review or act on.

TEST

A useful test: Imagine observing the user for 30 minutes. What would you actually see them reading, deciding, comparing, writing, organizing, or repeating?

Brainstorm three possible project problems.

For each idea, name the user and the task—not the app you might build.

Type your response here.

Which problem will you use for this homework, and why?

Choose the one you understand best and can test with realistic examples.

Type your response here.

2

Define the User and the Problem

Be specific enough that you could find and interview this person.

A. Primary user

Describe a role or situation—not “everyone.” Helpful details may include the user’s experience level, context, goal, and constraints.

Who is the primary user?

Example: “A university student preparing for a closed-book exam using lecture notes and readings.”

Type your response here.

B. The job they are trying to complete**What is the user trying to accomplish?**

Use an action: prepare, compare, decide, summarize, plan, draft, organize, evaluate, or monitor.

Type your response here.

C. The difficulty

What makes the task difficult, slow, repetitive, or frustrating?

Describe the current pain—not the absence of an AI tool.

Type your response here.

D. Evidence that the problem is real

What have you experienced or observed that suggests this problem matters?

Use a concrete example, short story, observation, or conversation.

Type your response here.

NOTE

Avoid invented certainty: You may begin with a hypothesis. Write “I believe...” or “I have observed...” when you have not yet interviewed users or collected evidence.

3

Map the Current Workflow

Understand the work before deciding what Claude should do.

Describe how the user completes the task today—even if the process is messy. This reveals where AI may help and where human judgment must remain.

What starts the current process?

What event, request, file, question, or deadline triggers the work?

Type your response here.

What steps does the user complete today?

Write 3–7 steps in order. Include tools, searches, decisions, and communication.

Type your response here.

Where does the user spend the most time or make the most mistakes?

Identify one or two bottlenecks.

Type your response here.

What parts require human judgment, context, or approval?

Examples: choosing priorities, verifying facts, making sensitive decisions, or sending final communication.

Type your response here.

Current workflow summary

Step	Who / tool	What happens	Pain point or risk
1	Type here	Type here	Type here
2	Type here	Type here	Type here
3	Type here	Type here	Type here
4	Type here	Type here	Type here
5	Type here	Type here	Type here
6	Type here	Type here	Type here

4

Design the First AI Workflow

Define what Claude receives, does, produces, and leaves to the human.

A. Inputs**What information will the user provide?**

Identify required and optional inputs. Examples: files, notes, criteria, links, preferences, or questions.

Type your response here.

Provide one realistic example input.

Use a real or fictional example that resembles what the project will receive.

Type your response here.

B. Claude's work**What should Claude do with the input?**

Describe the reasoning process as steps: extract, classify, compare, research, calculate, draft, verify, or ask questions.

Type your response here.

C. Output

What should Claude produce?

Specify the format and sections. The output should help the user make a decision or complete an action.

Type your response here.

Provide one realistic example output or output outline.

A structured outline is acceptable if the complete output would be long.

Type your response here.

D. Human control

What should the user review, decide, approve, or edit?

Do not automate sensitive or high-impact actions without an explicit review point.

Type your response here.

5

Set Scope and Define Success

A smaller, testable project is stronger than a large, vague promise.

A. First-version scope**What will the first version do?**

List the minimum capabilities needed to solve the core problem.

Type your response here.

What will the first version NOT do?

Examples: no automatic sending, no live database, no mobile app, no long-term memory, or no high-stakes recommendations.

Type your response here.

B. Success measures

Choose two or three measures you can evaluate during the project. Avoid vague goals such as “make it good.”

Measure	Example question
Accuracy	Are the claims supported by the provided information?
Completeness	Did the output include every required section?
Usefulness	Would the user act differently or save meaningful effort?
Time saved	How long does the workflow take compared with the current process?
Consistency	Does the same type of input receive a similarly structured result?
Trust	Does the workflow clearly show uncertainty and request review when needed?

How will you know the project is successful?

Write 2–3 measurable criteria and, where possible, a target.

Type your response here.

C. Assumptions and risks**What assumptions are you making?**

Examples: users have clean notes, the source material is accurate, or the task is completed in English.

Type your response here.

What could go wrong?

Consider missing information, hallucinations, privacy, bias, unclear instructions, or over-reliance on the result.

Type your response here.

6

Write the One-Sentence Definition

This sentence should make your project understandable immediately.

This project helps **[specific user]** complete **[specific task]** by turning **[input]** into **[output]**,
reducing **[time, effort, errors, or confusion]**.

Write your one-sentence project definition.

Read it aloud. A classmate should understand it without extra explanation.

Type your response here.

Completed Example: Study Pack Builder

Project brief field	Example response
Primary user	A university student preparing for a closed-book exam using lecture notes and assigned readings.
Problem	The student has a large amount of material but struggles to identify the most important concepts and test their understanding efficiently.
Current process	The student rereads notes, highlights material, writes a summary, and manually creates a few practice questions.
Input	Lecture notes, reading excerpts, course learning objectives, and the exam date.
Claude workflow	Organize concepts by learning objective; identify definitions, relationships, and likely misconceptions; ask for missing context; create a study guide and practice questions; label anything not supported by the source material.
Output	A structured study guide, a 15-question quiz with an answer key, and a list of weak areas to review.
Human role	The student reviews the study guide, confirms course relevance, answers the quiz, and decides what to study next.
Success	At least 90% of factual claims are traceable to the source material; the output follows the requested format; the student reports saving at least 30 minutes.
Not in version 1	No grade prediction, no automatic access to the learning platform, and no claim that generated questions will match the real exam.

Final Project Brief — Submission Copy

Use the answers from the previous pages to create a clear, concise brief below. Aim for one page of substantive content.

PROJECT NAME — Choose a simple working title.

Type your response here.

PRIMARY USER — Who is the main user, and in what situation will they use this?

Type your response here.

PROBLEM — What task are they trying to complete, and what makes it difficult?

Type your response here.

CURRENT WORKFLOW — How is the task completed today? Summarize the main steps and bottleneck.

Type your response here.

PROPOSED AI WORKFLOW — What inputs will Claude receive, what will it do, and what output will it produce?

Type your response here.

HUMAN REVIEW — What must the user verify, decide, edit, or approve?

Type your response here.

SUCCESS MEASURES — List 2–3 criteria you will use to evaluate the first version.

Type your response here.

FIRST-VERSION NON-GOALS — What will this version intentionally not do?

Type your response here.

ONE-SENTENCE DEFINITION — Use the sentence structure from Section 6.

Type your response here.

Submission Checklist

Done	Submission check
☐	My project has one primary user and one main problem.
☐	I described the current workflow before proposing the AI workflow.
☐	I identified the required inputs and the expected output.
☐	I included one realistic example input and one realistic example output or output outline.
☐	I explained what Claude should do step by step.
☐	I included at least one human review or approval point.
☐	I wrote 2–3 measurable success criteria.
☐	I clearly stated what the first version will not do.
☐	My one-sentence definition is specific and easy to understand.
☐	I reviewed Claude-generated suggestions and made the final decisions myself.

How This Homework Will Be Reviewed

Criterion	What strong work demonstrates
Specificity	The user, problem, input, and output are clearly defined.
Problem understanding	The brief explains the current process and the real bottleneck.
Workflow logic	Claude's role is appropriate, structured, and connected to the problem.
Scope	The first version is realistic and includes clear non-goals.
Human judgment	Important review, approval, privacy, and uncertainty needs are recognized.
Testability	Success can be evaluated using realistic examples and measurable criteria.
Clarity	Another person can understand the project without additional explanation.